Name: _____

WORKSHEET

Proper divisors

1 What is the difference between the factors of a number and the proper divisors of a number?

2 List the proper divisors of the following numbers.

a 72 _____

b 125 _____

c 48 _____

d 200 _____

3 The larger a number is, the harder it is to list its proper divisors. The easiest way to find all proper divisors is first express a number as a product of its prime factors. Consider the number 360.

a Express 360 as a product of its prime factors, using either your CAS calculator or finding them manually.

The factors of any number can be written using all the different combinations of its prime factors e.g. 2, $2^3 \times 3$, etc.

b List the different combinations of the prime factors. *Hint*: there are 23 different combinations.

c Evaluate each of the combinations, and then use these to write the list of proper divisors for 360.

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4 List the proper divisors for each of these numbers:

a 945

b 1701

c 5445

d 630

e 1890

f 67375

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Answers

1 The factors of a number are all numbers that divide evenly into the original number, including 1 and itself. The proper divisors of a number are all numbers that divide evenly into the original number, *excluding* itself.

2 **a** 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36

b 1, 5, 25

c 1, 2, 3, 4, 6, 8, 12, 16, 24

d 1, 2, 4, 5, 8, 10, 20, 25, 40, 50, 100

3 **a** $2^3 \times 3 \times 5$

b $2, 2^2, 2^3, 3, 3^2, 5, 2 \times 3, 2 \times 3^2, 2 \times 5, 2 \times 3 \times 5, 2 \times 3^2 \times 5, 2^2 \times 3, 2^2 \times 3^2, 2^2 \times 5, 2^2 \times 3 \times 5, 2^2 \times 3^2 \times 5, 2^3 \times 3, 2^3 \times 3^2, 2^3 \times 5, 2^3 \times 3 \times 5, 2^3 \times 3^2 \times 5, 3 \times 5, 3^2 \times 5.$

c 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72, 90, 120, 180

4 **a** 1, 3, 5, 7, 9, 15, 21, 27, 35, 45, 63, 105, 135, 189, 315

b 1, 3, 7, 9, 21, 27, 63, 81, 189, 243, 567

c 1, 3, 5, 9, 11, 15, 33, 45, 55, 99, 121, 165, 363, 495, 605, 1089, 1815

d 1, 2, 3, 5, 6, 7, 9, 10, 14, 15, 18, 21, 30, 35, 42, 45, 63, 70, 90, 105, 126, 210, 315

e 1, 2, 3, 5, 6, 7, 9, 10, 14, 15, 18, 21, 27, 30, 35, 42, 45, 54, 63, 70, 90, 105, 126, 135, 189, 210, 270, 315, 378, 630, 945

f 1, 5, 7, 11, 25, 35, 49, 55, 77, 125, 175, 245, 275, 385, 539, 875, 1225, 1375, 1925, 2695, 6125, 9625, 13475

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